

Mechanical force transmission across myofascial chains

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Introduction

The skeletal muscles cannot be considered as independent actuators (see Chapter 12). Instead, due to their intimate connections to the adjacent soft tissue, they may be influenced by forces originating from neighboring or even distant body locations. Some authors have refined this idea, suggesting the existence of highly specified mechanical pathways: it is assumed that myofascial chains transmit particularly large amounts of force owing to the in-series arrangement of their components (Figure 13.1). If this were true, not only the active component of the locomotor system but also the connective tissues linking the incorporated skeletal muscles (e.g. tendons, ligaments or the deep fascia) would have to possess higher force transmission capacities in the longitudinal rather than in the transversal direction.

Basic prerequisites and assumptions of myofascial force transmission

A quick view into the micro-anatomy of the connective tissue apparently supports the above hypothesis. Tendons and ligaments exhibit almost parallel collagen fibers oriented in one direction, which enables them to withstand and transfer high elongating forces. The collagen of the deep fascia has a lattice-like arrangement with a fiber-crossing angle of 80 degrees (Benetazzo et al., 2011). This construction allows for a flexible adaptation to both stretching and contraction of the underlying muscle.

Nevertheless, a particularly high longitudinal strength, similar to tendons and ligaments, can be expected because the fibers are rather oriented towards the muscle's line of pull. Indeed, findings from histological and biomechanical studies confirm the anisotropic features of the deep fascia. The fascia lata, which ensheathes the muscles of the thigh, has similar material properties to the patellar tendon (Butler et al., 1984). Its maximal stiffness is up to 7-fold higher in the longitudinal (283 Newton per millimeter) than in the transversal plane (41.9 Newton per millimeter) (Otsuka et al., 2018). Stress tests of tissues from other body regions which compared the maximal resilience in both directions showed that the loading capacity in the longitudinal plane is 5-fold higher for the crural fascia (Eng et al., 2014; Stecco et al., 2014) and 3-fold higher for the fasciae of the abdominal wall (Kirilova et al., 2011). In summary, these data show that not only ligaments and tendons, but also the deep fasciae, are stronger strain transmitters in the longitudinal direction, which often corresponds to the primary direction of muscle force development.

The mere existence of myofascial chains, as well as a high longitudinal loading capacity of the tissues connecting the included skeletal muscle components, do not necessarily imply functional relevance per se. Only if relevant forces can be transmitted across the continuities, are exercise and treatment according to them justifiable.

From a mechanical perspective, local alterations of tissue stiffness are a prerequisite of force

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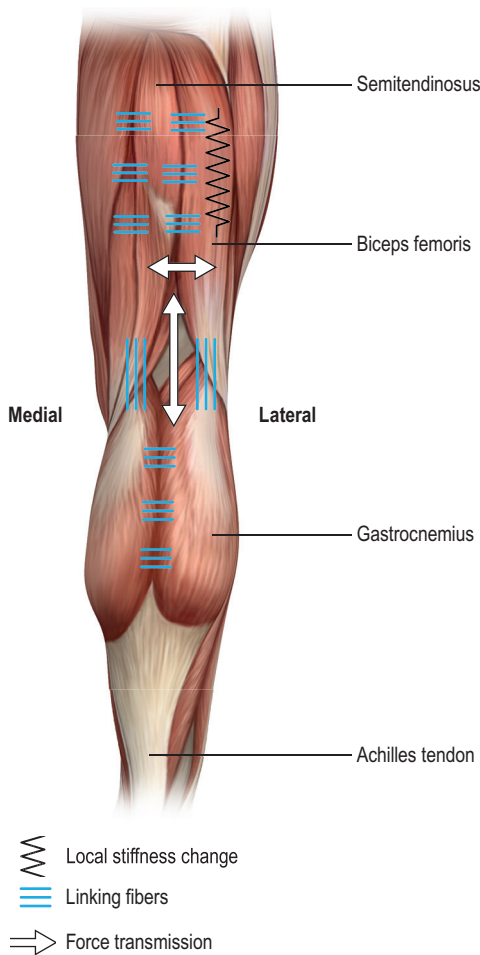


Figure 13.1 Schematic illustration of myofascial force transmission in the lower leg

Due to direct tissue continuity, local forces produced in the biceps femoris may be transferred to muscles located in parallel (e.g. the semitendinosus) and to muscles arranged in-series (e.g. the gastrocnemius). Concepts of myofascial chains assume higher degrees of force transmission between serially connected muscles.

transmission to other structures. They can occur via different mechanisms (Figure 13.2): as a first option, the muscle may represent the origin. Both acute contractions and chronic hypertonicity increase its diameter and, doing so, stiffen the surrounding fascia. According to mathematical

models, the developing radial stress is substantial, amounting to ~50% of the longitudinal stress (Findley et al., 2015). Otsuka et al. (2019) used ultrasound elastography (Chapter 20) to quantify fascial stiffness changes upon muscle contraction. Isometric knee extension at 60% of maximal voluntary strength led to large alterations of the connective tissue properties: compared with the resting state, the stiffness of the fascia lata was ~2.4-fold higher over both the rectus femoris muscle and the vastus lateralis muscle. While these increments were registered for the longitudinal direction, much lower (but still significant) contraction-induced stiffness increases occurred in the transversal plane of the fascia, which further solidifies the above evidence for the higher strain transmission capacity in the direction of the muscle's line of pull. In addition to the passive interaction through muscle expansion, mutual interactions with the surrounding connective tissue also occur, due to a special architectural feature. Some muscles present direct fibrous expansions into the fascia, which allow a selective tensioning of the connective tissue. This can, for instance, be observed in the latissimus dorsi, whose muscular fibers tighten the triceps brachii fascia (Stecco et al., 2008), and also in the pectoralis major muscle, whose fibers reach into the brachial fascia of the arm (Stecco et al., 2009).

In addition to being tensioned by the skeletal muscle, the fascia has two of its own mechanisms to regulate its stiffness. Several studies have demonstrated the occurrence of myofibroblasts, an intermediate contractile cell type between fibroblasts (known from wound healing) and smooth muscle cells in the fasciae of the limbs (e.g. the gastrocnemius, the fascia lata and the plantar aponeurosis) and the lumbar fascia of the back (Bhattacharya et al., 2010; Schleip et al., 2019). It can be assumed that myofibroblast contraction, which seems to be governed by the

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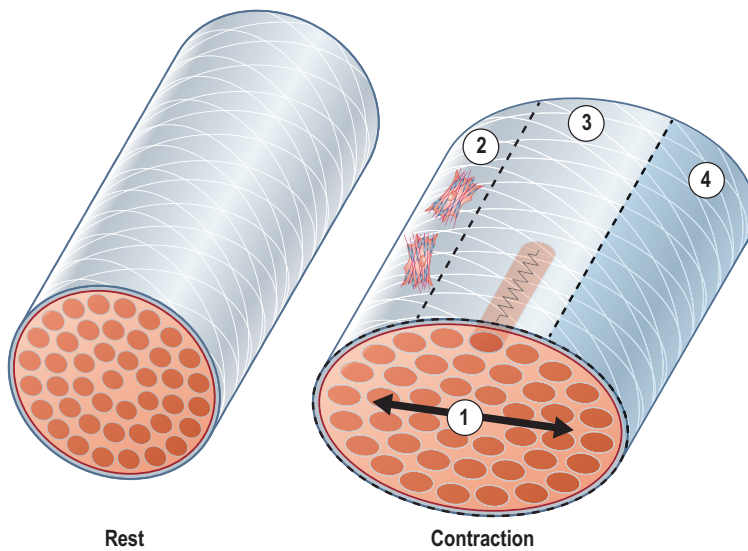


Figure 13.2 Schematic representation of four pathways modifying local fascia stiffness

Muscle contraction (right) changes its volume, transmitting radial stress to the extramuscular connective tissue (1). At the same time, direct insertions of muscle fibers allow tensioning of specific parts of the fascia (3). Without muscle contribution, fascial stiffness can increase due to myofibroblast activity (2) and changes in the water content (4).

vegetative nervous system (Schleip et al., 2019), creates a basal fascial tension in the long term, which may be excessively increased in some cases. The second mechanism, which explains short-term rises of stiffness, consists of altered matrix hydration, which has been shown to manifest following stretching.

Load me up Scotty: What do cadaveric studies and laboratory experiments tell us about force transmission?

In view of the particularly high resilience of fascia in the longitudinal direction and the variety of mechanisms by which its stiffness can be modified, researchers have made extensive efforts to quantify the degree of in-series myofascial force transmission. Typically, related experiments were carried out in dissected cadavers, applying traction to one chain component and measuring the resulting strain or morphological displacement in a connected neighboring structure. Most available work focused on the posterior longitudinal myofascial line extending

between the plantar foot and the neck (plantar aponeurosis, Achilles tendon, gastrocnemius, hamstrings, sacrotuberous ligament, lumbar fascia, erector spinae).

- Two studies (Carlson et al., 2000; Erdemir et al., 2004) indicate a substantial degree of force transmission between the plantar aponeurosis and the Achilles tendon. Up to 50% of the forces measured in the tendon can be transmitted to the aponeurosis during simulated walking (Erdemir et al., 2004).
- One study (Cruz-Montecinos et al., 2015) has investigated the mechanical behavior of the continuity between the gastrocnemius and the hamstrings. Upon pelvic motion (which stretches the dorsal thigh muscles), a cranial displacement of the gastrocnemius fascia is visible.
- The findings of three trials (van Wingerden et al., 1993; Vleeming et al., 1989; 1995) point towards mechanical interactions between the hamstrings and the lumbar spine region. Van Wingerden et al. (1993) and Vleeming et al. (1989) showed traction of the biceps

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femoris to tense the sacrotuberous ligament (which connects to the lumbar fascia), although the strength of this effect was highly variable. Vleeming et al. (1995) did not measure forces but found that the pull of the biceps femoris muscle led to local movement in the thoracolumbar fascia.

In addition to the posterior longitudinal line, force transmission has also been verified for some parts of the posterior diagonal chain (latissimus dorsi, lumbar fascia, contralateral gluteus maximus, vastus lateralis).

- Two studies (Barker et al., 2004; Vleeming et al., 1995) applied traction to the latissimus dorsi and gluteus maximus muscles and observed significant displacements in the lumbar fascia, which, to some extent, were also visible on the contralateral body side. One study (Carvalhais et al., 2013) performed dynamic stretching of the latissimus muscle and observed an altered resting position as well as an increase in passive stiffness of the opposite hip.
- Hitherto, the mechanical relevance of the continuity between the gluteus maximus and the vastus lateralis of the quadriceps femoris muscle has not been investigated.

Finally, initial evidence is available for the anterior diagonal chain, which includes the adductor longus, rectus abdominis, as well as the contralateral pectoralis major muscle.

- One trial (Norton-Old et al., 2013) has investigated the mechanical interaction between the adductor longus and the rectus abdominis. The authors of the study found that force applied to the adductor muscle leads to small and highly variable deformations of the rectus sheath.
- To date, no evidence is available for the continuity between the rectus abdominis and the pectoralis major muscle.

Despite the plethora of studies pointing towards the occurrence of in-series myofascial force transmission, several factors limit the validity of their findings. Above all, as identified, the vast majority of the experiments were conducted with cadavers. If not using fresh-frozen specimens, dissection is frequently performed on embalmed bodies. Fixation in formalin, a typically used agent in cadaveric research, causes the formation of collagenous cross-links and a reduction in the concentration of hyaluronic acid (Abe et al., 2003; Lin et al., 1997). Both processes may render the tissue more sticky and, together with the changes in the muscle fiber pennation angle also observed (Martin et al., 2001), reduce the amount of mechanical force transmitted. Notwithstanding, it cannot be concluded that the effects measured in cadaveric experiments necessarily underestimate the true magnitude: in the living organism, muscle activity can support but also counteract force transmission.

Now you see me: Visualizing myofascial force transmission may open new frontiers in research

In view of the limitations of cadaveric biomechanical testing, recent studies have attempted to translate the early findings into *in vivo* settings. In a pioneering but explorative study, Cruz-Montecinos et al. (2015) tested the potential interaction between two major components of the posterior longitudinal chain: the hamstrings and the gastrocnemius muscle. The authors instructed their participants to perform repeated pelvic motions while simultaneously assessing the position of the gastrocnemius fascia using live ultrasound. In sitting position and with the knees extended, anterior pelvic tilt (mean maximum: 6.6 degrees) caused small cranial displacements of the fascia of ~1.5 mm. In an even more standardized subsequent trial conducted by our

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workgroup (Wilke et al., 2020), which additionally used electromyography to control for confounding muscle activity, similar findings were made. Upon passive motion of the upper ankle joint, substantial movements of the medial hamstrings can be detected: the more the foot approaches end-range dorsal extension, the more the semitendinosus muscle and its fascia are displaced into the caudal direction. Both trials can be regarded as the first *in vivo* investigations proving in-series myofascial force transmission in humans. Further research needs to elucidate if the effects found in the lower limb occur to a similar degree in the trunk and the upper limb.

Remote exercise effects *in vivo*: Magic or a proof of myofascial force transmission?

Compelling evidence suggests that a local treatment can evoke functional changes at distant locations pertaining to the same myofascial chain. Such remote exercise effects have particularly been described for the posterior longitudinal chain. Self-myofascial release (see Chapter 27) is a popular method that is assumed to affect the stiffness of the targeted tissue. Following an intensive rolling massage of the plantar aponeurosis, sit and reach testing revealed clinically relevant increases in hamstring extensibility, which may be the result of a mechanical force transmission (Grieve et al., 2012; Wilke et al., 2019). However, the treatment response displays considerable variation. While young individuals mostly achieve a clinically relevant improvement of hamstring extensibility, trivial or no effects are more likely to occur with increasing age (Wilke et al., 2019). This can possibly be explained by the progressive loss of tissue continuity between the plantar aponeurosis and the Achilles tendon (see Chapter 12), which significantly reduces the amount of transmittable force.

Stretching is another popular intervention, which has been used to study non-local treatment effects. After three 30-s static stretching bouts of the gastrocnemius and the hamstring muscles, Wilke et al. (2016) observed a remote increase in sagittal cervical spine mobility. Again, it can be hypothesized that this effect evolves from the continuity between the targeted leg muscles and the neck extensor muscles. In a follow-up experiment, the same authors (Wilke et al., 2017) found the treatment to be as effective as local stretching of the neck muscles. Furthermore, range of motion was also examined in the frontal (lateral flexion) and transversal (rotation) plane. Interestingly, increases of flexibility not only affected the sagittal plane but also the other two movement directions. As the neck extensors (being the cranial part of the long dorsal myofascial chain) primarily limit flexion-extension movements, this may suggest that force transmission does not occur predominantly in-series. Irrespective of this, the finding of non-inferiority of remote stretching could be highly relevant in the treatment of unspecific neck pain, which is often associated with restricted range of motion. As local treatments, for example spine manipulation, can lead to severe adverse effects such as cervical arterial dissection (Puentedura et al., 2012), non-local interventions represent a safer, viable alternative.

Like cadaveric research, *in vivo* studies on myofascial force transmission also have some limitations that require further consideration. The most important of these is that the vast majority of the available trials used functional outcomes, which are an excellent way of simulating realistic training/therapeutic settings. However, without the direct measurement or visualization (and with this a proof) of force transmission, it cannot be inferred that the stretching or self-myofascial release

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interventions definitely caused the observed increases in range of motion. In addition to the connective tissue, peripheral nerves also cross multiple joints, and could mechanically affect distant body locations when stretched. Furthermore, it is well established that stretching has both mechanical and neuronal effects as it increases stretch tolerance. It is possible that this supraspinal reaction is not specific to the treated joint but leads to a general change in the stretch perception threshold. Despite encouraging findings, there is, hence, an urgent need for additional studies examining the cause-effect relationship of interventions based on myofascial chains. In essence, it is crucial to combine (1) the proof-of-principle imaging approaches used by Cruz-Montecinos et al. (2015) and our research group and (2) the application of a treatment in a clinically realistic scenario.

Treatments of musculoskeletal disorders based on myofascial chains

Frequently, therapists, coaches and other health professionals working with the concept of myofascial chains apply remote exercise treatments aiming to resolve musculoskeletal disorders at distant body locations. Although the preceding sections of this chapter have highlighted some open questions, initial evidence appears to support this approach.

Plantar fasciitis (also called fasciosis because inflammation is not always present) is a common cause of heel pain and may be affected by non-local abnormalities in the posterior longitudinal chain. In fact, patients with plantar fasciitis exhibit increases in the stiffness of the gastrocnemius and even the hamstrings (Bolívar et al., 2013; Harty et al., 2005; Labowitz et al., 2011). Retrospective analyses of surgical treatments revealed that the release of the proximal medial gastrocnemius is superior to local fasciotomy

(Monteagudo et al., 2013). Before surgery, stretching is a popular treatment among practitioners. According to a survey with therapists in the UK, more than half of the respondents always include non-local exercise for the hamstrings and/or the gastrocnemius into their routine treatment. However, Garrett and Neibert (2013) performed a literature review examining the efficacy of this approach. Their analysis of four studies showed that calf-stretching as a stand-alone treatment is not superior to a comparative or sham treatment.

Clinical example

Myofascial continuity can be easily experienced with a simple experiment. In standing, lift one leg, flexing the knee by ~60 to 90 degrees. Now move your foot between plantar flexion and dorsal extension. This should not be difficult and your ankle should display a large range of motion (ROM). Now extend the knee and repeat the same movement. Probably the ROM has decreased a bit, which can be ascribed to the bi-articularity of the gastrocnemius (knee extension pre-stretches the muscle as its origin is at the femur bone). In a third step, position your leg on a chair or a desk. While keeping your knee straight, test your ankle ROM. Again, it should have decreased slightly. If you now flex your hip (lower your upper body towards the leg on the table without flexing the lumbar spine), a further reduction will occur and in some cases, no more movement is possible (do not cheat by moving your pelvis back and forth or by flexing your toes!). What has happened? All the described steps have successively tightened the posterior longitudinal chain, which in the end restricted movement in the ankle.

Besides plantar fasciitis, individuals suffering from low back pain (LBP) could benefit from a treatment based on the posterior longitudinal chain. While it is unclear if this is always due to higher stiffness, patients with an unspecific disorder exhibited decreased hamstring extensibility (Fasuyi et al., 2017; Halbertsma et al., 2001;

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Tafazzoli and Lamontagne, 1996). In a prospective study with adolescents, tight hamstrings were shown to predict the occurrence of LBP to a small degree (Feldman et al., 2001). To date, trials assessing the effectiveness of hamstring stretching to alleviate LBP are sparse. Franca et al. (2012) compared a 6-week treatment (two weekly sessions of 30 minutes) based on the posterior longitudinal chain (stretching of the calf, hamstrings and erector spinae) with segmental stabilization exercises. After the intervention period, both groups had reduced their pain level and improved physical function, but the differences were more pronounced for segmental stabilization. Additionally, there was no inactive control group and no group that performed remote stretching only.

Non-local abnormalities of back pain patients do also affect the posterior diagonal chain. Joseph et al. (2014) recruited a sample of individuals with sacroiliac joint dysfunction. Using ultrasonography, they determined the resting position of the humeral head on the body side contralateral to symptom localization. Compared with an age- and sex-matched control group, the patients exhibited a significantly increased anterior translation of the humeral head, which may indicate an insufficiency of the latissimus dorsi muscle. As the latter is connected to the contralateral gluteus maximus via the thoracolumbar fascia, clinicians and therapists should be aware that shoulder issues and back pain may be linked or caused by each other. Other studies examining the potential relevance of the posterior diagonal chain in back pain used electromyographic muscle activity as an outcome. When performing a unilateral hip extension in the prone position, women with chronic LBP displayed increased firing of the ipsilateral gluteus maximus and the contralateral latissimus muscle (Kim et al., 2014). In a similar way, during walking, participants with sacroiliac

joint dysfunction demonstrated altered activity of the gluteus maximus on the involved side and increased activity of the opposite latissimus dorsi. After performing a strengthening torso rotation program, physiological conditions were restored (Mooney et al., 2001). While this finding is intriguing, two issues remain debatable. First, it has not yet been investigated if the electromyographic changes within the muscle-fascia lines are due to a force transmission effect and, second, regarding the study of Mooney et al. (2001), it is unclear if their corrective exercises also induced an alleviation of the back pain symptoms.

Finally, groin pain represents another pathology that may be treated by means of non-local exercise. Its prevalence is particularly high in football (soccer) players, causing pain during running, changes of direction and kicking motions. Both the adductor longus and the rectus abdominis muscle can be affected (Valent et al., 2010). As they are directly connected within the anterior diagonal chain, this observation is not surprising. However, while a combined treatment of both muscles seems prudent, to date, no study has examined its effectivity.

What to do next: an outlook

Mechanical force transmission across myofascial chains certainly represents one of the most promising fields of movement-related connective tissue research. It seems plausible that local dysfunctions (e.g. pain or restricted range of motion) have an impact on remote tissues, as changes of the mechanical tissue properties may radiate via the muscle-fascia lines. However, it has to be underlined that (1) the current state of evidence is indeed sufficient to justify a multidimensional approach in the design of exercise programs, which may include the use of non-local exercise interventions as an adjunct, but (2) a paucity of research exists regarding their effectiveness as a stand-alone treatment.

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